

NAME

mbROUTETIME – Program to output a list of the times when a survey passes the waypoints of a planned survey route.

VERSION

Version 5.0

SYNOPSIS

mbROUTETIME **--route-file=routefile** **{-Rroutefile}** **[--format=format** **{-Fformat}]** **[--input=file** **{-Ifile}]** **[--output=waypointtimefile** **{-Owaypointtimefile}]** **[--range-threshold=rangethreshold** **{-Urangethreshold}]** **[--verbose** **{-V}]** **[--help** **{-H}]**

DESCRIPTION

mbROUTETIME –

MB-SYSTEM AUTHORSHIP

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OPTIONS

--format=format **{-Fformat}**

Sets the format for the input swath data using **MBIO** integer format identifiers. This program uses the **MBIO** library and will read any swath format supported by **MBIO**. A list of the swath data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *format* = 0.

--help **{-H}**

This "help" flag cause the program to print out a description of its operation and then exit immediately.

--input=filename **{-Ifilename}**

Sets the input filename. If *format* > 0 (set with the **-F** option) then the swath data contained in *infile* is read and processed. If *format* < 0, then *infile* is assumed to be an ascii file containing a list of the input swath data files to be processed and their formats. The program will read the data in each one of these files. In the *infile* file, each data file should be followed by a data format identifier, e.g.:

```
datafile1 11
datafile2 24
```

This program uses the **MBIO** library and will read any swath format supported by **MBIO**. A list of the swath data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *infile* = "datalist.mb-1".

--output=waypointtimefile **{-Owaypointtimefile}**

Sets the name of the output file listing, for each route waypoint reached, the longitude, latitude, and time (in decimal epoch seconds) at which the survey passed that waypoint. Default: *waypointtimefile* is the input file or datalist name (set with the **-I** option) with the suffix "_wpttime_d.txt" appended.

--route-file=routefile {-Rroutefile}

Sets the name of the input **MBgrdviz** route file containing the planned survey waypoints against which the survey data are checked. This option is required.

--range-threshold=rangethreshold {-Urangethreshold}

Sets the distance in meters that the survey navigation must come within a route waypoint for that waypoint to be considered reached. Default: *rangethreshold* = 25.0.

--verbose {-V}

The **-V** option causes **mbrouetime** to print out status messages.

EXAMPLES

Suppose you are going to run the MBARI Mapping AUV on the Coaxial Segment of the Juan de Fuca Ridge. You have created a route file called Coaxial2009_1v3.rte using **mbgrdviz**. In order to create an MBARI AUV mission script for mission beginning with a spiral descent to an altitude of 50 m, followed by a survey run at a 75 m altitude, the following will suffice:

```
mbrouetime -I Coaxial2009_1v3.rte          -A75/30/75 -B1          -G0 -MSL -N50
-R220/83/0.3          -O Coaxial2009_1v3.cfg          -S1.5 -L30 -W100 -V
```

The resulting mission script has the following header:

```
# This MBARI Mapping AUV mission file has been generated
# by the MB-System program mbrouetime run by
# user <caress> on cpu <shepard> at <Thu Jul 30 11:36:47 PDT 2009>
#
# Mission Summary:
#   Route File:      Coaxial2009_1v3.rte
#   Mission File:    Coaxial2009_1v3.cfg
#   Distance:        79501.503455 (m)
#   Estimated Time:   57203 (s) 15.890 (hr)
#   Abort Time:       60067 (s)
#   Max battery life: 64800 (s)
#   Safety margin:    1800 (s)
#   Ascend time:      2932 (s)
#   Way Points:       44
#   Route Points:     550
#   Survey behavior:   WaypointDepth
#   Descent style:     Spiral descent
#   Mapping sonar control enabled:
#       Multibeam enabled
#       Multibeam receive gain:      83
#       Multibeam transmit gain:     220
#       Multibeam minimum range fraction: 0.3
#       Subbottom enabled
#       Low sidescan enabled
#       High sidescan disabled
#
# Mission Parameters:
#   Vehicle Speed:      1.500000 (m/s) 2.915769 (knots)
#   Desired Vehicle Altitude: 75 (m)
#   Minimum Vehicle Altitude: 75 (m)
#   Abort Vehicle Altitude: 30 (m)
#   Maximum Vehicle Depth: 2525.307922 (m)
#   Abort Vehicle Depth: 2562.807922 (m)
#   Descent Vehicle Depth: 3 (m)
```

```
# Spiral descent depth: 2324.917643 m
# Spiral descent altitude: 50 m
# Forward Looking Distance: (m)
# Waypoint Spacing: 100 (m)
# GPS Duration: 600 (s)
# Descend Rate: 0.417 (m/s)
# Ascend Rate: 1 (m/s)
# Initial descend Duration: 300 (s)
# Setpoint Duration: 30 (s)
#
# The primary waypoints from the route file are:
# <number> <longitude (deg)> <latitude (deg)> <topography (m)> <distance (m)> <type>
# 0 -129.588618 46.504590 -2384.917643 0.000000 3
# 1 -129.583151 46.507559 -2412.977865 533.709482 3
# 2 -129.569223 46.503420 -2548.389974 1697.143568 1
# 3 -129.566359 46.501080 -2494.963053 2037.557099 3
# 4 -129.548611 46.529852 -2539.510864 5512.537193 4
# 5 -129.551250 46.530628 -2562.807922 5732.537193 3
# 6 -129.568962 46.501924 -2531.034424 9199.497998 4
# 7 -129.571600 46.502699 -2519.138489 9419.497998 3
# 8 -129.553889 46.531404 -2547.114624 12886.458803 4
# 9 -129.556529 46.532180 -2514.533569 13106.458803 3
# 10 -129.574238 46.503475 -2470.815735 16573.419607 4
# 11 -129.576876 46.504250 -2444.596313 16793.419607 3
# 12 -129.559168 46.532956 -2521.781921 20260.380412 4
# 13 -129.561807 46.533732 -2537.382141 20480.380412 3
# 14 -129.579514 46.505026 -2429.459961 23947.341216 4
# 15 -129.582152 46.505801 -2412.764343 24167.341217 3
# 16 -129.564447 46.534508 -2545.397705 27634.302021 4
# 17 -129.567086 46.535284 -2534.068665 27854.302021 3
# 18 -129.584791 46.506576 -2398.283020 31321.262826 4
# 19 -129.587429 46.507352 -2390.671509 31541.262826 3
# 20 -129.569726 46.536059 -2489.889282 35008.223630 4
# 21 -129.572365 46.536835 -2465.280823 35228.223630 3
# 22 -129.590068 46.508127 -2389.067017 38695.184435 4
# 23 -129.592706 46.508902 -2409.290771 38915.184435 3
# 24 -129.575005 46.537610 -2466.533142 42382.145240 4
# 25 -129.577645 46.538386 -2491.371094 42602.145240 3
# 26 -129.595345 46.509677 -2397.609253 46069.106044 4
# 27 -129.597984 46.510452 -2413.315918 46289.106044 3
# 28 -129.580285 46.539161 -2499.048889 49756.066849 4
# 29 -129.582925 46.539937 -2523.030640 49976.066849 3
# 30 -129.600622 46.511227 -2443.481018 53443.027653 4
# 31 -129.603261 46.512002 -2419.008240 53663.027653 3
# 32 -129.585565 46.540712 -2518.522400 57129.988458 4
# 33 -129.588205 46.541487 -2498.521301 57349.988458 3
# 34 -129.605900 46.512777 -2450.386536 60816.949263 4
# 35 -129.608539 46.513552 -2473.623230 61036.949263 3
# 36 -129.590845 46.542262 -2487.422180 64503.910067 4
# 37 -129.593485 46.543038 -2491.040466 64723.910067 3
# 38 -129.611178 46.514327 -2472.610657 68190.870872 3
# 39 -129.606711 46.517999 -2485.114583 68723.704236 3
# 40 -129.557338 46.509809 -2488.398743 72619.152031 3
```

```

# 41 -129.550415 46.521262 -2545.285828 73998.189601 3
# 42 -129.600724 46.536447 -2470.920736 78209.372536 3
# 43 -129.606972 46.525648 -2443.988281 79501.503455 4
#
# A total of 550 mission points have been defined.
#
# Define Mission parameters:
#define MISSION_SPEED 1.500000
#define MISSION_DISTANCE 79501.503455
#define MISSION_TIME 57203
#define MISSION_TIMEOUT 60067
#define DEPTH_MAX 2525.307922
#define DEPTH_ABORT 2562.807922
#define ALTITUDE_DESIRE 75.000000
#define ALTITUDE_MIN 75.000000
#define ALTITUDE_ABORT 30.000000
#define GPS_DURATION 600
#define DESCENT_DEPTH 3.000000
#define SPIRAL_DESCENT_DEPTH 2324.917643
#define SPIRAL_DESCENT_ALTITUDE 50.000000
#define DESCEND_DURATION 300
#define SETPOINT_DURATION 30
#define GPSMINHITS 10
#define ASCENDRUDDER 3.000000
#define ASCENDPITCH 45.000000
#define ASCENDENDDEPTH 2.000000
#define DESCENDRUDDER 3.000000
#define DESCENDPITCH -30.000000
#define MAXCROSSTRACKERROR 30
#define RESON_DURATION 6
#q

```

SEE ALSO

mbsystem(1), mbgrdviz(1)

BUGS

Perhaps.